

Mohit

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Profile

Robotic Simulation Engineer with 2+ years of experience in BIW plant simulations and digital twin development, specializing in Siemens Process Simulate and NVIDIA Omniverse (OpenUSD). Proven expertise in robotic simulation, OLP, and virtual plant integration, combined with 3 years of leadership experience as CTO in a startup. Focused on building scalable digital manufacturing systems for Industry 5.0 driving Physical AI.

Work Experience

Robotic Simulation Engineer, EDAG PS INDIA, Gurgaon, Haryana

06/2024 – Present

Digital Twin & Omniverse Development | 01/2026 – Present

- Developed Python-based Omniverse Kit extensions for interactive digital twin environments.
- Built USD-based modular workflows for scalable virtual plant integration.
- Implemented Gaussian Splatting pipelines for 3D reconstruction and immersive visualization.

BIW Plant Automation (As per BMW standards) | 06/2024 – 01/2026

- Designed BIW production cells (fixtures, frames, layouts) for automotive manufacturing.
- Performed robotic simulations using Siemens Process Simulate (cycle time, collision validation).
- Developed and validated Offline Programming (OLP) with optimized robot path planning.

Founder and Ex-CTO, Ground Robotics Pvt. Ltd., AIC, IIT Delhi

01/2021 – 06/2024

Government-funded Innovation Projects | 10+ national recognitions

- **GVIC – Smart Irrigation System (2024):** LoRa-based flood irrigation system for canal farming; funded under NIDHI PRAYAS (upto 10 lakh) and incubated at AIC IIT Delhi.
- **SWAN – Water Monitoring System (2022–2023):** IoT-based water quality monitoring device for aquaculture; funded under RUSA 2.0 (upto 6 lakh), recognized across national platforms.

Additive Manufacturing analysis | FDM 3D Printing

- **Mechanical Analysis of 3D Printing Materials (2023):** Evaluated 50+ thermoplastics across FDM, SLA, DLP, and SLS processes for performance and suitability
- **FDM Dimensional Accuracy Study (2022):** Tested PLA, Tough PLA, TPU 95A, and ABS on Ultimaker S5; contributed to design validation and documentation standards

Education

B.Tech. Mechanical Engg., GJUS&T, Hisar, Haryana – NBA accredited course

2020-24

(Gold Medallist and first to achieve B.Tech. Hons. Degree under Mechanical Department)

Intermediate, H.D. Public School, Rohtak, Haryana – CBSE Affiliated

2019-20

Matriculation, S.G.S.A.M.N.E.M. School, Indore, MP – CBSE Affiliated

2017-18

Internships & Trainings

CAD-CAM Training, MSME Technology Centre, Rohtak | On-site 09/2023 –11/2023

- Completed 200-hour training in CNC operations using Sinumerik EMCO WinNC (20+ programs).
- Gained hands-on exposure to machining processes: turning, milling, drilling, boring, grooving, and engraving.

Design and Research Intern, Exobot Dynamics Pvt. Ltd., AIC, IIT Delhi | On-site 06/2023 –07/2023

- Resolved precision issues in bionic hand (PIP joint), achieving tolerance up to 0.01 mm.
- Performed static & dynamic analysis and iterated designs for functional optimization.

Management Development Programme, MSME DFO, Karnal | On-site 11/2022 –12/2022

- Completed government-certified training in digital marketing strategies and online business growth.
- Gained exposure to SEO, social media marketing, branding, and digital campaign management

Mechanical Design Internship, PNT Robotics, Maharashtra | Remote 07/2022 –08/2022

- Developed and enhanced product user experience through detailed designs and interaction models.
- Applied SolidWorks for detailed designs, optimizing needs into product experiences.

SolidWorks Training, MSME Technology Centre, Rohtak | On-site 07/2022 –08/2022

- Rapidly modelled industrial designs by creating 50+ parametric engineering models.
- Collaborated with 10+ efficient design assemblies' workflows.

Extracurricular Activities

Founder and Ex-Captain, iConnect – PDUIIC, GJUS&T Hisar (Now **Mentor)** 10/2023 –02/2024

- Founded and scaled the university's entrepreneurship society, leading a 60+ member team (14 core).
- Organized the first Techfest (10+ events, 2000+ attendees), coordinating sponsors, judges, & administration.

Outcome: Now iConnect Society is established and scaled as the university's central entrepreneurship body by integrating Smart India Hackathon (internal selection process) and merging the Entrepreneurship Club (earlier under T&P Cell) into a unified, student-led innovation ecosystem; currently mentoring its growth.

NPTEL, SWAYAM | Initiative by Ministry of Human Resource Development 2021-2025

- NPTEL Enthusiast Certification (Jan-Apr 2023), NPTEL Stars, SWAYAM Portal
- All India Rank 5, Fundamental & Conceptual Design, NPTEL -IIT Madras
- Built a strong foundation in **robotics** through Robotics, IITKGP and Advanced Robotics, IIT Kanpur.

Other Certifications: **German - I, IITM** | Fundamental & Conceptual Design, IITM | Programming and DSA in Python, IITB | Understanding Incubation and Entrepreneurship, IITB | Introduction to IoT, IITKGP | The Joy of Computing using Python, IIT Ropar | Fundamentals of Artificial Intelligence, IITG

LinkedIn Learning, LinkedIn 2024-Present

Leadership Mindsets | Time Management Fundamentals | AI in Project Management | How to Win Any Negotiation with Chris Voss | Sharpen Your Logic | Think Faster, Talk Smarter

C Certified Cadet (Alpha Grade), National Cadet Corps, 3rd HR Battalion NCC 2021-2024

- Completed NCC Additional Course 3-year degree Programme | 3HR BNCC Approved (2021-24)
- Best Cadet Award, Annual Training Camp-199, NCC at SKPPS, Siwani, Haryana
- 1st Prize – Test of Elementary Training (Weapon Proficiency), ATC-110, NCC at E5, Hansi, Haryana

Sports & Fitness

- Karate Player (State Representative, Haryana) 2021-2024
- Cult Fit Member (Cult Ninja - Consistency Recognition) 2025-Present

Honours & Awards

- **Guest Speaker** – Student Startup Bootcamp, AIC, Shiv Nadar Institution of Eminence, Delhi NCR 06/2025
- **Finalist** – Build Your Own Business, Himalayan Startup Trek 2023, IIT Mandi, Himachal Pradesh 10/2023
- **Finalist** – State Level Hackathon, iHUB DivyaSampark@ IIT Roorkee, DCRUST, Murthal, Haryana 04/2023
- **Finalist** – Investors Vista event, E-summit'23, IIT Roorkee, Uttarakhand 02/2023
- **1 st Prize** – Ideastrome event, E-summit'23, IIT Roorkee, Uttarakhand 02/2023
- **Finalist** – Empresario'23, Global Business Model Competition, IIT Kharagpur, West Bengal 02/2023
- **1 st Prize** – B-plan event, Techsprdha'23 & E-summit'23, NIT, Kurukshetra, Haryana 01/2023
- **Most Impactful Idea** – Moonshot event, Prometeo'23, IIT Jodhpur, Rajasthan 01/2023
- **1 st Prize** –Business Darbar event, Prometeo'23, IIT Jodhpur, Rajasthan 01/2023
- **Student of the Year**, Udaan Fest 2K22, GJUS&T, Hisar, Haryana 05/2022
- **1 st Prize** – Young leaders' event, Techspardha'22, NIT, Kurukshetra, Haryana 04/2022
- **1 st Prize** – SSSS category, 4th Technovation Hackathon & Innovation Budge, Sharda University, Delhi 02/2022

Certifications: (Short-term Skill Enhancement Programmes)

- Problem Solving using Python, HackerRank 12/2025
- AI Mastermind Programme – Outskill Platform 09/2025
- C-Certified Cadet (Alpha Grade), National Cadet Corps, P H HP & C Directorate 06/2024
- Barclays LifeSkills Programme (Recruitment Training), GTT Foundation, Pune, Maharashtra 10/2023
- MATLAB and Simulink Onramp, MathWorks Training Services 08/2023
- Python Programming with Data Structure & Algorithm, YBI Foundation 07/2023
- Robotic Process Automation Developer, UiPath - Business Automation Platform 05/2023
- Aptitude Training Program, TIME Institute, Hisar, Haryana 01/2023

Key Proficiencies

Core Competencies (Advanced):

- **Simulation & Automation:** Siemens Process Simulate (BIW Simulation, OLP, validation)
- **CAD Design:** SolidWorks (3D modelling & assemblies), CATIA (2D drafting)
- **Digital Twin Development:** NVIDIA Omniverse (OpenUSD, Kit, Python workflows)
- **Additive Manufacturing** (3D printing): Fused Deposition Modelling, Stereolithography

Other Technical Tools: MS Office, Canva (Graphic Design), Python Programming, Arduino, IoT, UiPath, DELMIA, ANSYS Workbench, PLC Concepts (Siemens TIA), ENCO WinNC (Sinumerik CNC), MATLAB, AutoCAD, Fusion360, Siemens UG NX

Soft Skills:

- Team Management
- Problem Solving
- Strategic Planning
- Leadership
- Decision Making
- Research & Analysis

Languages

- Hindi - Native proficiency
- English - Professional proficiency
- Punjabi - Limited proficiency
- German - Elementary Proficiency
- Japanese - Elementary Proficiency